Yixiao Zhang

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Education

Queen Mary University of London 2020 – 2024
PhD in Artificial Intelligence and Music *London, the United Kingdom*

- Thesis: Improving Controllability and Editability for Pretrained Text-to-Music Generation Models
- Advisors: Prof. Simon Dixon & Dr. Mark Levy

University of Electronic Science and Technology of China 2015 – 2019
BEng. in Computer Science, GPA: 3.74/4 *Chengdu, Sichuan, China*

Research Interests

Speech and audio generative AI; multimodal learning across text, audio, and visuals; foundation models for audio understanding and generation; diffusion models.

Work Experience

ByteDance Seed Feb 2025 – Present
Research Scientist in Foundation Model, Music Core Machine Learning Graduates - 2024 Start (PhD) *San Jose, USA*

- Serve as the primary contributor on **music understanding** for the Seed-Music foundation model, including DeepChorus-2 / DeepChorus-3 music structure analysis models (WASPAA 2025) and **AudioLLM-based** (MoE-30B) multi-lingual lyrics transcription and captioning systems providing large-scale supervision signals for pre-training data curation; performed **continual training and SFT** on this model, managing **distributed training up to 128 GPUs** using **FSDP2/DDP** with mixed-precision training.
- Developed Orpheus, the Seed-Music **evaluation framework**, integrating and continuously iterating metrics for text-to-music generation and editing models, and aligning them with research and product needs; developed interpretable **aesthetic evaluation pipelines** to identify high-musicality, high-fidelity training data, improving **large-scale data curation** efficiency and downstream model quality for production systems.
- Lead **post-training and music domain adaptation** of **SAM-Audio**, a **3B-parameter DiT-based** generative source separation model, for controllable music editing (**in-painting, style transfer, stem manipulation**); collaborate with the AR model team on training recipe design and ablation studies for music foundation models ranging from **0.7B to MoE-30B** parameters.

Internships

ByteDance Seed Oct 2024 – Jan 2025
Research Intern *Shanghai, China*

- Improved the performance of deep learning-based music structure analysis models, directly enhancing the quality of training data for the Seed-Music foundation model.

Stability AI Apr 2024 – Oct 2024
Research Intern *London, UK*

- Developed a Dreambooth-like finetuning method for Stable Audio Open, enabling generative model personalization via audio prompts. (Mentor: Dr. Jordi Pons & Dr. Julian Parker)

Sony AI Oct 2023 – Apr 2024
Research Intern *Tokyo, Japan*

- Conducted novel research on zero-shot music editing, leading to two first-author publications at top-tier conferences (IJCAI 2024, ISMIR 2025). (Mentor: Yukara Ikemiya)

Yamaha Corporation Jun 2023 – Sep 2023
Research Intern *Hamamatsu, Japan*

- Pioneered research into the potential of LLMs as Music AI Agents for controllable and iterative music generation. (Mentor: Dr. Akira Maezawa)

Microsoft Research Asia (MSRA) Oct 2018 – Jun 2019
Research Intern *Beijing, China*

- Conducted theoretical research on deep reinforcement learning algorithms and their convergence properties. (Mentor: Dr. Weiqing Liu & Dr. Xiao Yang)

Selected Publications

1. **Temporal Adaptation of Pre-trained Foundation Models for Music Structure Analysis** WASPAA 2025
   Y. Zhang, *Haonan Chen, Ju-Chiang Wang, Jitong Chen*
2. **Unlocking Text-to-Music Editing for Music Language Models via Instruction Tuning** ISMIR 2025
   Y. Zhang, *Y. Ikemiya, W. Choi, N. Murata, M. Martínez, W. Liao, L. Lin, G. Xia, Y. Mitsufuji, S. Dixon*
3. **MusicMagus: Zero-Shot Text-to-Music Editing via Diffusion Models** IJCAI 2024
   Y. Zhang, *Y. Ikemiya, G. Xia, N. Murata, M. Martínez, W. Liao, Y. Mitsufuji, S. Dixon*
4. **Steerable Long-term Music Audio Generation and Editing via Content-based Controls** IJCAI 2024
   Liwei Lin, Gus Xia, ***Yixiao Zhang****, Junyan Jiang*
5. **Content-based Controls For Music Large Language Modeling** ISMIR 2024
   Liwei Lin, Gus Xia, Junyan Jiang, ***Yixiao Zhang***
6. **Interpreting Song Lyrics With An Audio-Informed Pre-Trained Language Model** ISMIR 2022
   Yixiao Zhang, *Junyan Jiang, Gus Xia, Simon Dixon (Best Paper Award Nomination)*

Awards & Honors

Best Paper Award Nomination ISMIR 2022
For the paper "Interpreting Song Lyrics With An Audio-Informed Pre-Trained Language Model"

Professional Activities

Committee Member Roles
Organizing Committee, MIREX (2024, 2025) & CMMR (2025); Technical Committee, AES (2023, 2024)

Peer Reviewer / Program Committee
NeurIPS, ICML, ICLR, IJCAI, ISMIR, ICASSP, WASPAA, CHI, ACM Multimedia